2.1

) $\lim_{(x,y)\to(0,1)} (x+y) = 1$

$|x+y-1| < \varepsilon$

$|x+y-1| \leq |x| + |y-1| < \varepsilon$

sabemos que:

$0 < \left|\sqrt{x^2+(y-1)^2}\right| < \delta$

$\Rightarrow \sqrt{x^2+(y-1)^2} < \delta$

$\Rightarrow |x| < \delta \wedge |y-1| < \delta \Leftrightarrow |x| + |y-1| < 2\delta$

$\Rightarrow \varepsilon \leq 2\delta$

$\Rightarrow \delta = \dfrac{\varepsilon}{2}$

-//ion

$\lim_{(x,y)\to(0,0)} \dfrac{y^2}{\sqrt{x^2+y^2}} = 0$?

$\forall_{\varepsilon>0} \exists_{\delta>0}$ tal que $\forall_x \; 0 < |x-a| < \delta \Rightarrow |f(x) - \ell| < \varepsilon$

$\sqrt{x^2+y^2} < \delta$

$\left|\dfrac{y^2}{\sqrt{x^2+y^2}}\right| < \varepsilon \rightarrow \dfrac{y^2}{\sqrt{x^2+y^2}} \leq \dfrac{x^2+y^2}{\sqrt{x^2+y^2}} = \sqrt{x^2+y^2} < \delta$

$\Rightarrow \delta = \varepsilon$